

Simple Linear Regression: Influential Points, Interpretations

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MATH 240

Week 4, Lecture #3 — Ch. 7

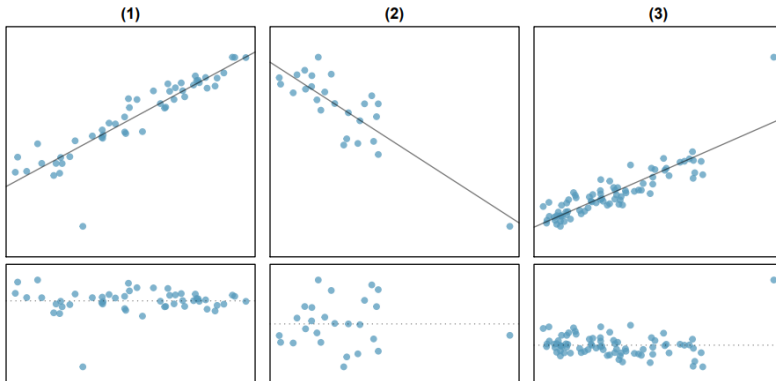
Objectives

- **Interpret** coefficient estimates
- **Make** predictions using new values of x with the fitted model

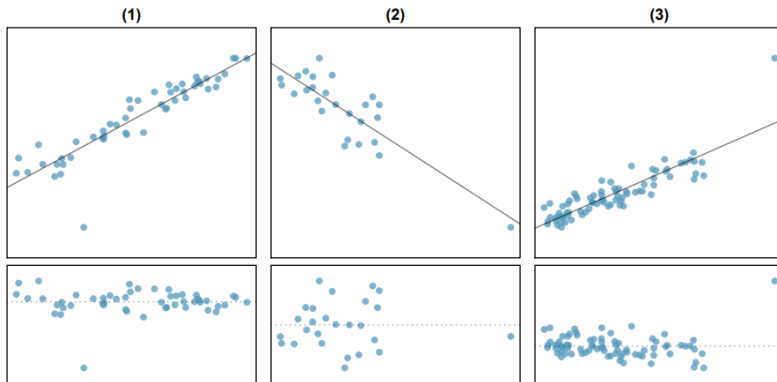
Sidenote: Influential Points

- **Influential point:** data points that affect the estimated coefficients of the least-squares line
 - ▶ When removed, the revised least-squares line will better fit the data
 - ▶ These points are generally far away from the least-squares line

Example: Understanding Influential Points

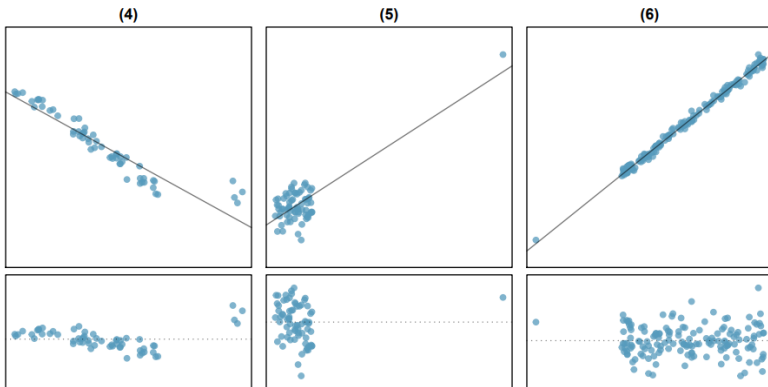


Example: Understanding Influential Points

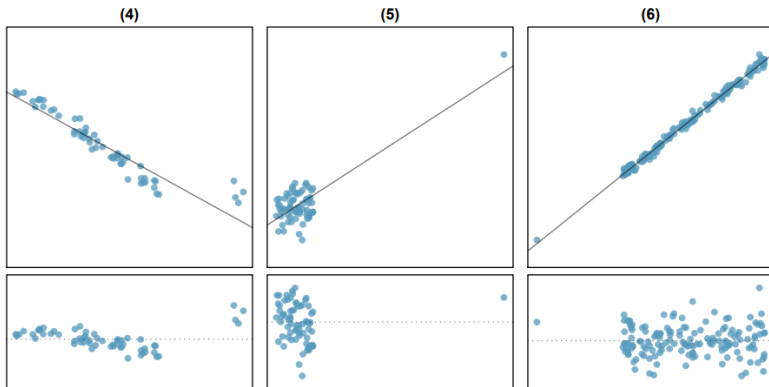


- ① One outlier – slightly influences the line
- ② One outlier – very close to the regression line, not very influential
- ③ One outlier – appears to pull the regression line up

Example: Understanding Influential Points



Example: Understanding Influential Points



- ④ A small cloud of 4 outliers, pulling the regression line up
- ⑤ One outlier, not very influential
- ⑥ One outlier, not very influential

Activity: Diagnostics with the mtcars Dataset

Activity: Comb through the diagnostic plots of the model you created from the previous activity. Make sure to detail what you see in the four plots, as well as justify if the conditions are satisfied:

- **Linearity**
- **Nearly normal residuals**
 - ▶ Outliers?
 - ▶ Influential points?
- **Constant variability**
- **Independent observations**

4. Interpretations

- Identify which β coefficients are significant (i.e. hypothesis testing)
 - ▶ Need to see if the β coefficients are significantly different from zero
 - ▶ For covariates: if significant, the covariate has an impact on the response variable
- Interpret the coefficients with context in mind

4. Interpretations - Summary

Using the following code,

- `summary(mod)`

Recall the following results:

```
Call:
lm(formula = mtcars$mpg ~ mtcars$displ)

Residuals:
    Min       1Q   Median       3Q      Max
-4.8922 -2.2022 -0.9631  1.6272  7.2305

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 29.599855   1.229720  24.070 < 2e-16 ***
mtcars$displ -0.041215   0.004712  -8.747 9.38e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.251 on 30 degrees of freedom
Multiple R-squared:  0.7183,    Adjusted R-squared:  0.709
F-statistic: 76.51 on 1 and 30 DF,  p-value: 9.38e-10
```

4. Interpretations - Summary Breakdown

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	$\hat{\beta}_0$	$SE(\hat{\beta}_0)$	$T_{\hat{\beta}_0}$	p-value $_{\hat{\beta}_0}$
Variable	$\hat{\beta}_1$	$SE(\hat{\beta}_1)$	$T_{\hat{\beta}_1}$	p-value $_{\hat{\beta}_1}$

- **Estimate:** estimate of the β coefficients
- **Std. Error:** standard error associated with the β coefficients

$$SE(\hat{\beta}_0) = \sqrt{MSE \left(\frac{1}{n} + \frac{\bar{x}^2}{SS_x} \right)} \quad SE(\hat{\beta}_1) = \sqrt{\frac{MSE}{SS_x}}$$

where

- ▶ $SS_x = \sum (x_i - \bar{x})^2$ for $i = 1, \dots, n$
- ▶ MSE is the sum of squared errors divided by $n - 2$
 - ★ e.g. $MSE = SSE/(n - 2)$, seen in previous slide

4. Interpretations - Summary Breakdown

	Estimate	Std. Error	t value	$\Pr(> t)$
(Intercept)	$\hat{\beta}_0$	$SE(\hat{\beta}_0)$	$T_{\hat{\beta}_0}$	p-value $_{\hat{\beta}_0}$
Variable	$\hat{\beta}_1$	$SE(\hat{\beta}_1)$	$T_{\hat{\beta}_1}$	p-value $_{\hat{\beta}_1}$

- **t value:** test statistic associated with β coefficients

$$T = \frac{\hat{\beta}_i}{SE(\hat{\beta}_i)} \sim t(df = n - 2) \text{ under } H_0$$

- **$\Pr(>|t|)$:** p-value associated with β coefficients (e.g. $P(t > |T|)$)

4. Interpretations - Summary Breakdown

Residual standard error: 3.251 on 30 degrees of freedom
Multiple R-squared: 0.7183, Adjusted R-squared: 0.709
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- **Residual standard error:** square root of mean squared error
- **Degrees of freedom:** $n - 2$
- **Multiple R^2 :** proportion of variation in response variables explained by covariates
- **Adjusted R^2 :** adjusted proportion of variation in response variables explained by covariates
- **F statistic:** used to check if there are any variables that are significant in the model

4. Interpretations - Summary Breakdown

Formulas from previous slide (note: k is number of covariates in model):

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$df = n - 2$$

$$\text{Multiple } R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

$$\text{Adjusted } R^2 = 1 - \frac{\left(\frac{SSE}{n-k}\right)}{\left(\frac{SST}{n-1}\right)}$$

4. Interpretations - Hypothesis Testing for Coefficients

For defining a hypothesis test for some regression coefficient, we expect there to be **no change or difference**.

Thus, for some regression coefficient to test, we have the following hypotheses:

$$H_0 : \beta_i = 0 \text{ vs. } H_a : \beta_i \neq 0$$

- We are testing to see if there is enough evidence to see that a variable is significant in explaining the response.
- We generally will focus our attention on the slope, not the intercept coefficient

4. Interpretations - Conclusions for Hypothesis Testing

Thus, we can use the following justifications to reject H_0 :

- $p\text{-value} < \alpha$
- $|T| > |t_{1-\alpha/2, n-2}|$
- 0 not in confidence interval:
 $(\hat{\beta}_i - t_{1-\alpha/2, n-2} \times SE(\hat{\beta}_i), \hat{\beta}_i + t_{1-\alpha/2, n-2} \times SE(\hat{\beta}_i))$

If we reject H_0 , there is significant evidence that the **coefficient is different from zero**. For the slope coefficient, this results in a **significant variable**.

Theory/Tutorial: Interpretation with the mtcars Dataset

Recall the following results:

```
Call:
lm(formula = mtcars$mpg ~ mtcars$disp)

Residuals:
    Min       1Q   Median       3Q      Max
-4.8922 -2.2022 -0.9631  1.6272  7.2305

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  29.599855   1.229720  24.070  < 2e-16 ***
mtcars$disp  -0.041215   0.004712  -8.747  9.38e-10 ***
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Residual standard error: 3.251 on 30 degrees of freedom
Multiple R-squared:  0.7183,    Adjusted R-squared:  0.709
F-statistic: 76.51 on 1 and 30 DF,  p-value: 9.38e-10
```

Theory/Tutorial: Interpretation with the mtcars Dataset

We can calculate the test statistics as such:

$$\begin{aligned}T(\hat{\beta}_0) &= \frac{\hat{\beta}_0}{SE(\hat{\beta}_0)} \\&= \frac{29.599855}{1.229720} \\&= 24.070\end{aligned}$$

$$\begin{aligned}T(\hat{\beta}_1) &= \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \\&= \frac{-0.041215}{0.004712} \\&= -8.747\end{aligned}$$

We can get the p -values as such:

$$\begin{aligned}p\text{-value}_{\hat{\beta}_0} &= 2 * pt(24.070, 30, lower.tail = F) \\&= 3.5783 \times 10^{-21}\end{aligned}$$

$$\begin{aligned}p\text{-value}_{\hat{\beta}_1} &= 2 * pt(8.747, 30, lower.tail = F) \\&= 9.38 \times 10^{-10}\end{aligned}$$

Theory/Tutorial: Interpretation with the mtcars Dataset

Conclusions

Since the following p -values are less than a significance level of 0.05, as seen below:

$$\begin{aligned}p\text{-value}_{\hat{\beta}_0} &= 2 * pt(24.070, 30, lower.tail = F) \\ &= 3.5783 \times 10^{-21}\end{aligned}$$

$$\begin{aligned}p\text{-value}_{\hat{\beta}_1} &= 2 * pt(8.747, 30, lower.tail = F) \\ &= 9.38 \times 10^{-10}\end{aligned}$$

We can reject the null hypothesis for both coefficients and say the following:

- The intercept is significantly different from 0
- The slope is significantly different from 0
 - ▶ The displacement of a vehicle is a significant covariate for explaining MPG

Theory/Tutorial: Interpretation with the mtcars Dataset

Confidence Interval

We can also calculate the confidence interval for our coefficients through R using the following code:

- `confint(mod)`

	2.5 %	97.5 %
(Intercept)	27.0884	32.1113
disp	-0.0508	-0.0316

As such, we are 95% confident that the true value of β_1 is contained between -0.05084 and -0.03159.

Activity: Interpretation with Your Model

Activity: With the model results from R,

- Determine if the β coefficients are significant from 0.

5. Predictions

- We may want to predict what y would be given some x that we do not have real data for
- Useful for getting a good idea about what to expect, but not a precise estimate
- Be careful of extrapolation! Values of x not within domain of original data may result in questionable or nonsensical predictions

Tutorial: Prediction with the mtcars Dataset

Suppose we have a vector of x values that we want to estimate y for. In R, we can use the following function to do this task:

- `predict(model, data)`

Note that the x values must be placed in a single-column data frame, for which the column is named exactly as the x variable we use.

- `new_data <- data.frame(var_name = c())`

Tutorial: Prediction with the mtcars Dataset

Say we want to find predictions for the following x values: 100, 153, 254, 164, 374, 275, 166, 250. We can use the following code to get predicted responses for these x values:

- `predict(model, data)`

Note that the x values must be placed in a single-column data frame, for which the column is named exactly as the x variable we use.

- `new_data <- data.frame(displacement = c(100, 153, 254, 164, 374, 275, 166, 250))`
- `predict(mod, new_data)`

	1	2	3	4	5	6	7	8
1	25.48	23.29	19.13	22.84	14.19	18.27	22.76	19.30

Tutorial: Prediction with the mtcars Dataset

Activity: Choose 10 values of your explanatory variable x that you want to generate predictions of the response variable y for, using the model you have created in previous activities.

Quick Aside: Categorical Predictors

- Most of what we have talked about can be applied to categorical predictors
- Use of indicator variable to specify if coefficient is associated with that group
- For each variable with k groups, we need $k - 1$ indicator variables

Tutorial: Categorical Predictors

Suppose we want to see how the type of transmission (automatic or manual) affected the MPG of a car, using the mtcars dataset.

In some cases, variables in R may not be in a factor form. If appropriate (i.e. the variables are categorical), we can change the form of the variable to be of a factor form as seen below:

- `mtcars$am <- factor(mtcars$am)`

We can now fit the model as we did before:

- `mod2 <- lm(mpg ~ am, data = mtcars)`
- `summary(mod2)`

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.1474	1.1246	15.25	0.0000
am1	7.2449	1.7644	4.11	0.0003

Tutorial: Categorical Predictors

- `summary(mod2)`

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.1474	1.1246	15.25	0.0000
am1	7.2449	1.7644	4.11	0.0003

- Notice the 1 after 'am'. This specifies that the coefficient applies to group 1 (i.e. manual).
- In this case, having a manual will increase the MPG by 7.2449 on average, leading to an average MPG of 24.3923.
- There will always be a reference level. In this case, automatic is the reference level.
 - ▶ As such, having an automatic will result in an MPG of 17.1474 on average.